Good-Endpoint Variant of the Weighted Metric Trace (Route δ, Plan 2, Building Block GoodEndpointTrace)

Abstract

We strengthen the weighted metric trace of [1] by producing a Borel level $\widehat{\rho} \in G_I \cap G_\tau$ within $O(\delta_T)$ of $\rho_\delta = 1 - \kappa\sqrt{\delta_T}$ at which $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C\delta_T$. Here G_I is the Fusco–Julin good set $\{D_I \le \eta_I\}$ and $G_\tau = \{w(\rho) \ge \tau_0\}$ is the profile-gap good set. The point is that the boundary-layer transfer of [2] only requires the metric trace at *some* level $\widehat{\rho}$ with $\rho_\delta - \widehat{\rho} = O(\delta_T)$: the collar volume $|\Omega \setminus E_{\widehat{\rho}}| \le \omega_n n(\kappa\sqrt{\delta_T} + O(\delta_T))$ still has order $\sqrt{\delta_T}$, so squaring preserves the sharp exponent. This removes the need for $\widehat{\rho}$ to coincide with the exact endpoint ρ_δ. Honest status of the bad-level kinetic hypotheses is recorded in §6: both bad-I action (K_I) and bad-τ action (K_τ) remain load-bearing for the underlying trace theorem we invoke, and this variant does not by itself eliminate either.

1 Notation and standing assumptions

We follow [1]. Fix $n \ge 2$, $R \ge 2$, $\rho_* \in (0,1)$, $\kappa = \kappa(n, \rho_*) > 0$. Let $\Omega \subset B_R$ be open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $\delta := \delta_T(\Omega)$, $w(\rho) = -t'(\rho)$, $d\mu(\rho) = w(\rho)d\rho$, $F_\rho := \rho^{-1}E_\rho$. The smallness threshold from [1, (eq:delta-smallness)] gives

$$\delta \le \delta_0 := \left(\frac{1-\rho_*}{2\kappa}\right)^2, \qquad \rho_\delta := 1 - \kappa\sqrt{\delta} \ge \frac{1+\rho_*}{2}, \qquad L^* := \frac{\rho_\delta - \rho_*}{4} \ge \frac{1-\rho_*}{8}.$$

Set $J^* := [\rho_\delta - L^*, \rho_\delta]$, the order-one trace window. Define

$$\begin{aligned} G_I &:= \{\rho \in [\rho_*, \rho_\delta] : D_I(t(\rho)) \le \eta_I\}, & B_I &:= [\rho_*, \rho_\delta] \setminus G_I, & \\ G_\tau &:= \{\rho \in [\rho_*, \rho_\delta] : w(\rho) \ge \tau_0\}, & B_\tau &:= [\rho_*, \rho_\delta] \setminus G_\tau, & \tau_0 &:= \rho_*/(2n), \end{aligned}$$

where $\eta_I = \eta_I(n, R, \rho_*) > 0$ is fixed in [1, Hyp. 3.5].

2 The good-endpoint trace

Theorem 2.1 (Good-endpoint weighted metric trace)**.** *There exist constants $C = C(n, R, \rho_*, \kappa) > 0$, $C_1 = C_1(n, R, \rho_*, \kappa) > 0$, and a smallness threshold $\delta_0'(n, R, \rho_*, \kappa) > 0$ such that, under the conditional hypotheses of [1, Hyp. 3.5–3.7] (i.e. the inputs that yield Theorem 3.3 of that block), the following holds.*

For every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta = \delta_T(\Omega) \le \delta_0'$, there exists a Borel level

$$\widehat{\rho} \in [\rho_\delta - C_1\delta, \rho_\delta] \cap G_I \cap G_\tau \tag{1}$$

such that

$$d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C(n, R, \rho_*, \kappa)\, \delta. \tag{2}$$

The proof has two steps: a measure-theoretic non-emptiness statement for the candidate window, and a direct application of the existing trace-uniform theorem of [1].

3 Bad-level measure bounds in Lebesgue form

Lemma 3.1 (Bad-set Lebesgue measures)**.** *There is* $C_{\mathrm{bad}} = C_{\mathrm{bad}}(n, R, \rho_*) > 0$ *such that*

$$\mathcal{L}^1(B_\tau) \;\le\; C_{\mathrm{bad}}\,\delta, \tag{3}$$

$$\mathcal{L}^1(B_I \cap G_\tau) \;\le\; C_{\mathrm{bad}}\,\delta. \tag{4}$$

Proof. The first bound is exactly the profile-gap input [1, Hyp. 3.7, eq. (profile-gap)].

For the second, the level-set deficit identity [1, (DH-DI-int)] combined with the definition of G_I gives

$$\mu(B_I) \;\le\; \eta_I^{-1} \int_{\rho_*}^{\rho_\delta} D_I(t(\rho))\, d\mu(\rho) \;\le\; \frac{C(n,\rho_*)}{\eta_I}\,\delta.$$

On $B_I \cap G_\tau$, $w(\rho) \ge \tau_0 = \rho_*/(2n)$, so $\mu(B_I \cap G_\tau) \ge \tau_0\,\mathcal{L}^1(B_I \cap G_\tau)$. Hence

$$\mathcal{L}^1(B_I \cap G_\tau) \;\le\; \frac{\mu(B_I)}{\tau_0} \;\le\; \frac{2nC(n,\rho_*)}{\rho_*\,\eta_I}\,\delta.$$

Taking the maximum of the two constants gives C_{bad}. □

Lemma 3.2 (Non-empty good window near ρ_δ)**.** *Let* $C_1 := 3C_{\mathrm{bad}}$ *with* C_{bad} *as in Lemma 3.1. Assume* $\delta \le \delta_0'(n, R, \rho_*, \kappa) := \min\{\delta_0, L^*/(2C_1)\}$*. Then*

$$\mathcal{L}^1\big([\rho_\delta - C_1\delta, \rho_\delta] \cap G_I \cap G_\tau\big) \;\ge\; C_{\mathrm{bad}}\,\delta \;>\; 0. \tag{5}$$

In particular the set in (1) *is non-empty and has positive Lebesgue measure.*

Proof. Write $I_\delta := [\rho_\delta - C_1\delta, \rho_\delta]$. The condition $\delta \le L^*/(2C_1)$ ensures $C_1\delta \le L^*/2$, so $I_\delta \subset J^* \subset [\rho_*, \rho_\delta]$, in particular $\mathcal{L}^1(I_\delta) = C_1\delta$. By Lemma 3.1,

$$\mathcal{L}^1(I_\delta \cap B_\tau) \le C_{\mathrm{bad}}\delta, \qquad \mathcal{L}^1(I_\delta \cap B_I \cap G_\tau) \le \mathcal{L}^1(B_I \cap G_\tau) \le C_{\mathrm{bad}}\delta.$$

The complement satisfies

$$I_\delta \cap G_I \cap G_\tau \;=\; I_\delta \setminus \big(B_\tau \cup (B_I \cap G_\tau)\big),$$

and the two excluded sets are disjoint (the first sits in B_τ, the second in G_τ). Hence

$$\mathcal{L}^1(I_\delta \cap G_I \cap G_\tau) \;\ge\; C_1\delta - 2C_{\mathrm{bad}}\delta \;=\; (3C_{\mathrm{bad}} - 2C_{\mathrm{bad}})\delta \;=\; C_{\mathrm{bad}}\delta \;>\; 0.$$

□

Remark 3.3 (Borel measurability)**.** The set G_I is Borel because $\rho \mapsto D_I(t(\rho))$ is Borel (coarea on a Sobolev function); G_τ is Borel because $w = -t'$ is Borel. Hence the candidate set in (1) is Borel, and any specific Borel selector (e.g. the essential infimum of the set, which is Borel-measurable because the set has positive measure) provides a Borel-measurable choice of $\widehat{\rho}$.

4 Proof of Theorem 2.1

Proof. Let $\delta \le \delta_0'$. By Lemma 3.2, the candidate set in (1) has positive Lebesgue measure; choose any $\widehat{\rho} \in [\rho_\delta - C_1\delta, \rho_\delta] \cap G_I \cap G_\tau$ (Borel selection exists by Remark 3.3; the conclusion below does not depend on the selection rule). Since $C_1\delta \le L^*/2$, $\widehat{\rho} \in J^* = [\rho_\delta - L^*, \rho_\delta]$.

The trace-uniform theorem [1, Thm. 3.3, eq. (trace-uniform)] states

$$\sup_{\rho \in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \;\le\; C_*(n, R, \rho_*, \kappa)\,\delta.$$

Specialising to $\rho = \widehat{\rho} \in J^*$ gives (2) with $C := C_*$. □

5 Verification: downstream boundary-layer transfer absorbs $\widehat{\rho}$

The boundary-layer transfer of [2] is stated at ρ_δ but only uses the two ingredients (2) and the collar volume bound. We verify both survive the replacement $\rho_\delta \rightsquigarrow \widehat{\rho}$.

Lemma 5.1 (Collar volume at $\widehat{\rho}$)**.** *For $\widehat{\rho}$ as in Theorem 2.1,*

$$|\Omega \setminus E_{\widehat{\rho}}| \;\le\; \omega_n n\,(1-\widehat{\rho}) \;\le\; \omega_n n\big(\kappa\sqrt{\delta} + C_1\delta\big) \;\le\; C'_{\mathrm{lay}}\sqrt{\delta}, \tag{6}$$

where $C'_{\mathrm{lay}} := \omega_n n(\kappa + C_1\sqrt{\delta'_0})$.

Proof. Since $E_{\widehat{\rho}} \subset \Omega$ and $|E_{\widehat{\rho}}| = \omega_n \widehat{\rho}^n$, $|\Omega \setminus E_{\widehat{\rho}}| = \omega_n(1-\widehat{\rho}^n)$. The mean-value theorem on $\rho \mapsto 1-\rho^n$ over $[\widehat{\rho}, 1] \subset [\rho_*, 1]$ gives $1-\widehat{\rho}^n \le n(1-\widehat{\rho})$. From (1),

$$1-\widehat{\rho} \;=\; (1-\rho_\delta) + (\rho_\delta - \widehat{\rho}) \;\le\; \kappa\sqrt{\delta} + C_1\delta \;\le\; (\kappa + C_1\sqrt{\delta'_0})\sqrt{\delta}.$$

□

Corollary 5.2 (Bounded sharp Saint–Venant via $\widehat{\rho}$)**.** *Under the hypotheses of Theorem 2.1, there is $C_{\mathrm{SV}} = C_{\mathrm{SV}}(n, R, \rho_*, \kappa) > 0$ such that*

$$\mathcal{A}(\Omega)^2 \le C_{\mathrm{SV}}\,\delta. \tag{7}$$

Proof. By the superlevel transfer lemma [2, Lemma 4.2] applied with $E_t = E_{\widehat{\rho}}$ (valid because $E_{\widehat{\rho}} \subset \Omega$ and the lemma is purely combinatorial in the volume defect),

$$\mathcal{A}(\Omega) \;\le\; \mathcal{A}(E_{\widehat{\rho}}) + \frac{2}{\omega_n}|\Omega \setminus E_{\widehat{\rho}}|.$$

By (2) and [2, Rem. 2.4], $\mathcal{A}(E_{\widehat{\rho}}) \le \omega_n^{-1} d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)$, so $\mathcal{A}(E_{\widehat{\rho}})^2 \le \omega_n^{-2} C\,\delta$. By Lemma 5.1, $|\Omega \setminus E_{\widehat{\rho}}|^2 \le (C'_{\mathrm{lay}})^2\delta$. Squaring with $(a+b)^2 \le 2a^2 + 2b^2$,

$$\mathcal{A}(\Omega)^2 \;\le\; \frac{2C}{\omega_n^2}\delta + \frac{8(C'_{\mathrm{lay}})^2}{\omega_n^2}\,\delta \;=:\; C_{\mathrm{SV}}\,\delta.$$

The exponent 2 is preserved exactly because the $O(\delta)$ contribution from $\rho_\delta - \widehat{\rho}$ is dominated by the $O(\sqrt{\delta})$ collar coming from $1-\rho_\delta = \kappa\sqrt{\delta}$; the cross term $O(\delta \cdot \sqrt{\delta}) = O(\delta^{3/2})$ is absorbed by $O(\delta)$ for $\delta \le 1$. □

6 Status of bad-level kinetic hypotheses

This section is the honest analysis the task brief asks for. We record exactly which of the two bad-level kinetic action bounds of [1, Hyp. 3.6], namely

$$\int_{B_I} |\dot{F}_\rho|^2_{\mathcal{X}}\, d\mu(\rho) \le C\delta, \tag{K_I}$$

$$\int_{B_\tau} |\dot{F}_\rho|^2_{\mathcal{X}}\, d\rho \le C\delta, \tag{K_τ}$$

are still load-bearing under the proof above.

What the proof above actually uses. The proof of Theorem 2.1 invokes the trace-uniform estimate [1, Thm. 3.3] as a black box, hence it inherits *all* of that theorem's dependencies. In the chain of [1]:

- the integrated action bound (its Thm. 3.1) uses (K_I) on B_I;
- the Lebesgue kinetic bound (its Thm. 3.2) uses the integrated action bound on G_τ plus (K_τ) on B_τ;
- the trace-uniform statement (its Thm. 3.3) uses both of the above as (H1), (H4) in its abstract trace lemma.

Thus, taken at face value, the proof above does *not* remove either (K_I) or (K_τ).

Can a refined proof remove (K_τ)**?** A potential refinement, sketched in the brief, is to repeat the trace argument starting from a Markov point $\rho_0 \in J^* \cap G_\tau$ extracted from the integrated action (K_I)-only bound, and then propagate to $\widehat{\rho} \in J^* \cap G_I \cap G_\tau$ by kinetic Cauchy–Schwarz on $[\widehat{\rho}, \rho_0]$. The honest obstruction to dropping (K_τ) is:

> The interval $[\widehat{\rho}, \rho_0]$ may intersect B_τ, and the Cauchy–Schwarz step needs $\int_{[\widehat{\rho},\rho_0]} |\dot{F}_\rho|^2_{\mathcal{X}}\, d\rho < \infty$ with the right δ-scaling. On $[\widehat{\rho}, \rho_0] \cap G_\tau$ the bound follows from $w \geq \tau_0$ and the integrated $d\mu$-action; on $[\widehat{\rho}, \rho_0] \cap B_\tau$ *some* δ-scaling input is needed.

The bound $\mathcal{L}^1(B_\tau) \leq C_{\mathrm{bad}}\delta$ controls only the measure of B_τ, not the $L^2(d\rho)$ integral of $|\dot{F}_\rho|_{\mathcal{X}}$ over B_τ. Without (K_τ) or a substitute (e.g. a pointwise Lipschitz bound on $\rho \mapsto F_\rho$ in $\mathcal{X}$, which is exactly the discredited A0), this Cauchy–Schwarz step is not closed. Hence the present good-endpoint variant does not remove (K_τ) on its own.

Can a refined proof remove (K_I)**?** The bad-I action input is used to bound the integrated action $\int_{[\rho_*,\rho_\delta]}(d^2_{\mathcal{X}} + |\dot{F}|^2_{\mathcal{X}})\, d\mu \leq C\delta$ on B_I, which then feeds Markov to find $\rho_0 \in J^* \cap G_\tau$. An alternative route is to skip the integrated action entirely and extract ρ_0 from the pointwise good-level estimate $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C\, D_I(t(\rho))$ valid on G_I ([1, Hyp. 3.5]) combined with the integrated budget $\int D_I\, d\mu \leq C\delta$. By Markov on $d\mu|_{G_I \cap J^*}$ one obtains $\rho_0 \in G_I \cap J^*$ with $d_{\mathcal{X}}(F_{\rho_0}, B_1)^2 \leq K_1\delta$ *without using* (K_I). The catch is that one then still has to propagate this distance bound from ρ_0 to $\widehat{\rho}$, which puts us back at the same Cauchy–Schwarz obstruction on $[\widehat{\rho}, \rho_0] \cap B_\tau$.

In summary:

- Replacing the exact endpoint ρ_δ by $\widehat{\rho} \in G_I \cap G_\tau$ near ρ_δ is a clean structural simplification of the downstream interface (Corollary 5.2).
- It does *not* by itself eliminate either bad-level kinetic hypothesis. Both (K_I) and (K_τ) remain load-bearing for the trace-uniform theorem that we invoke.
- A genuine elimination of (K_τ) would require a non-trivial substitute for the Cauchy–Schwarz transport across B_τ (e.g., a quantitative AC bound for $\rho \mapsto F_\rho$ at bad-density levels), which is exactly the discredited A0; we do not provide one here.

7 Conditionality summary

Theorem 2.1 is conditional on the same external ingredients as [1, Thm. 3.3]:

- the level-set deficit identity [3],
- the conditional same-centre Fusco–Julin trace package [1, Hyp. 3.5],

- the conditional good-level metric upper gradient [1, Hyp. 3.6],
- the bad-level action inputs (K_I) and (K_τ),
- the profile-gap/Talenti density input [1, Hyp. 3.7].

No new conditional input is introduced. The variant produces a Borel $\widehat{\rho}$ in $G_I \cap G_\tau \cap [\rho_\delta - C_1\delta, \rho_\delta]$, which the boundary-layer transfer of [2] absorbs without loss of the sharp exponent 2, as shown in Corollary 5.2.

References

[1] Plan 2 building block WeightedMetricTrace. Integrated action, kinetic bound, and weighted metric trace at ρ_δ; see in particular Theorems 3.1–3.3, Hypotheses 3.5–3.7, and the abstract trace Theorem 4.1.

[2] Plan 2 building block BoundaryLayerTransfer. Endpoint metric trace to bounded sharp Saint–Venant via superlevel transfer; Theorem 3.1 (transfer at the level $\widehat{\rho}$) and Theorem 4.1 (small-δ_T Saint–Venant).

[3] Plan 2 building block LevelSetIdentity. Exact level-set deficit identity $\frac{2\delta_T}{|\Omega|} = \frac{1}{|\Omega|} \int (D_H + D_I)/(n^2 \omega_n^{2/n} m^{1-2/n})\, dt$ and the profile-gap measure estimate $\mathcal{L}^1(B_\tau) \le C\delta$.